



Unit 3 · Outcome 1 · Maintaining Biodiversity

The three categories of genes, species & ecosystems

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Biodiversity isn't one thing you count — it's the variety of all life, measured at three nested levels. Get these three straight and the whole Biodiversity unit clicks into place. Drag the model below to explore how genes sit inside species, and species inside ecosystems.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



— DEFINITION

02 What is biodiversity?

Biodiversity (biological diversity) is the variety of all living things on Earth — every organism, the genes they carry, and the ecosystems they form. The trap is to read "variety" as "how many animals." Biodiversity is richer than a head-count: scientists describe it at three interconnected levels.

Variety of genes (alleles) within a species or population.

Variety of ecosystems / habitats across a region.

◆ ANALOGY

Analogy · the library Genetic diversity = the different editions and translations of one book — same title, many variations. Species diversity = the number of different titles on the shelf. Ecosystem diversity = the number of different libraries across a city. Lose editions and a title gets fragile; lose titles and a library gets thin; lose libraries and whole subjects disappear.

Throughout this module we'll survey one place — Kestrel Ridge Reserve, a fictional box-ironbark woodland in central Victoria with a creek, a grassy woodland and a rocky ridge — and use real Victorian species as concrete examples.

— LEVEL ONE

03 Genetic diversity

The variety of genes (alleles) within a species or a population — the differences in heritable traits between individuals of the same species.

Think of all the eastern grey kangaroos in one mob: they share a species, but differ slightly in size, colour, immune-system genes and disease resistance. That spread of alleles is the population's genetic diversity.

- It's the raw material for evolution. Natural selection can only act on variation that already exists.
- A genetically diverse population is more likely to contain individuals that survive a new disease, heatwave or change — so the species can adapt.
- Low genetic diversity (from a small, isolated population or inbreeding) means fewer options when conditions change, raising extinction risk.

• NOTE

Victorian example. The Helmeted Honeyeater — a critically endangered bird found only in Victoria — survives as a tiny population with very low genetic diversity, which is exactly why conservation programs work hard to boost its gene pool. (More on threats in KK10.)

— LEVEL TWO

04 Species diversity

The variety of different species in an area. It has two components — don't stop at the first one.

How evenly individuals are spread across those species (relative abundance).

Species diversity = richness + evenness. Two surveys can record the same number of species, yet differ in diversity because one is dominated by a single species. We'll prove this in the Richness vs evenness model — and it's the single most common 3-mark trap in the exam.

◆ ANALOGY

Analogy · the fruit bowl Richness = how many types of fruit are in the bowl. Evenness = whether it's a balanced mix, or 46 apples and one of everything else. Same number of types, very different bowl.

— LEVEL THREE

05 Ecosystem diversity

The variety of ecosystems across a region — the range of different habitats, communities and the ecological processes that happen within them.

An ecosystem is a community of organisms plus their physical environment, interacting as a unit. Ecosystem diversity zooms right out: how many different ecosystems does a landscape contain?

- High ecosystem diversity: Victoria alone spans alpine ash forests, mallee, temperate wetlands, native grasslands and rocky coasts — each a distinct ecosystem.
- Low ecosystem diversity: a region cleared and converted to a single use — one crop, one plantation — collapses many habitats into one.
- At Kestrel Ridge, three ecosystems sit side by side: the creek wetland, the grassy woodland and the rocky ridge — that variety is ecosystem diversity.

• NOTE

Watch the wording. Ecosystem diversity is the variety of ecosystems — not the number of animals living in one. Counting organisms is species-level; counting habitats is ecosystem-level.

— INTERACTIVE

06 Diversity Explorer — one place, three lenses

Here's the same patch of Kestrel Ridge seen through each lens. Switch between the three levels and watch what "variety" means change. Drag to look around.

— THE CLASSIC TRAP

07 Richness vs evenness — proof in 3D

⚠ EXAM TRAPS

Two heathland plots at Kestrel Ridge, each with the same five species (same richness). The only difference is how the 50 individuals are spread. Flip Plot B between balanced and dominated and watch the diversity verdict change — richness never moves.

• NOTE

The takeaway. Equal richness does not mean equal diversity. The plot with higher evenness has the greater species diversity. This idea becomes a calculation in KK07 (Simpson's Index).

— CONNECT IT

09 One connected web

The three levels aren't separate boxes — they're nested and interdependent. Genes make up species; species make up ecosystems. So a change at one level ripples to the others:

- Lose genetic diversity → the species can't adapt → it weakens or dies out.
- Lose a species → the ecosystem loses a role (a pollinator, a predator, a food source) → the ecosystem degrades.
- Lose an ecosystem → habitat vanishes for many species at once → species and genetic diversity both fall.

That's why protecting a single threatened endemic species (the focus of this whole outcome) means caring about all three levels at once — not just keeping a few individuals alive.

— READ THE VERB

11 Command words — do what's asked

◆ COMMAND WORDS

The verb in the question tells you how much to write. Marks are lost by describing when asked to distinguish, or listing when asked to explain.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

— STRUCTURE

12 Structured answers · Point → Evidence → Link

A clean short-answer structure: make your Point, support it with Evidence / an example, then Link back to the question. Here it is on a real prompt — reveal each line to study the moves.

— EXAM PRACTICE

13 Practice questions

★ PRACTICE & SELF-MARK

Original VCE-style questions. Attempt each one first, then reveal the weak answer, the model answer, and the marking guide — and self-mark to track your score.

- 1 Definition: the variety of all living things / life (not merely a count of organisms).

- 1 Correctly naming all three categories: genetic, species and ecosystem. Naming only two does not earn this mark.
- 1 Genetic diversity correctly defined — variety of genes/alleles within a species/population.
- 1 Species diversity correctly defined — variety of different species in an area.
- 1 A valid example of each and an explicit contrast (within vs between / one species vs many).
- 1 Identifies that both plots have equal richness (5 species).
- 1 States species diversity depends on richness and evenness.
- 1 Correctly contrasts evenness — Plot A even, Plot B dominated by one species.
- 1 Correct conclusion (Plot A) justified by the higher evenness.
- 1 Correct, specific effect on genetic diversity.
- 1 Correct, specific effect on species diversity.
- 1 Correct, specific effect on ecosystem diversity. (A generic "it's bad" earns 0.)
- 2 Genetic-diversity reasoning: fewer alleles / raw material for evolution; reduced ability to adapt; inbreeding in small populations → raised extinction risk. (1 each, max 2)
- 3 Interdependence: levels are nested (genes→species→ecosystems); a species performs an ecological role; change at one level affects the others. (1 each, max 3)

— LOCK IT IN

14 Recap

Category	Variety of...	Scope	Kestrel Ridge example
Genetic	genes / alleles	within one species	disease-resistance genes varying among Helmeted Honeyeaters
Species	different species (richness + evenness)	across species in an area	40 bird species recorded in the woodland survey
Ecosystem	ecosystems / habitats	across a region	creek wetland + grassy woodland + rocky ridge

- Biodiversity = the variety of all living things, at three levels.
- Genetic = within a species · Species = across species (richness and evenness) · Ecosystem = across habitats.
- The levels are nested and interdependent — change at one ripples to the others.

U3·01 · KK01 · MAINTAINING BIODIVERSITY

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• NOTE

VCAA dot point. The definition and three categories of biodiversity — genetic, species and ecosystem diversity. This is the foundation the entire outcome is built on, and it shows up in nearly every exam.

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Biodiversity is the number of animals and plants in an area. The three types are big animals, small animals and plants." — defines it as a head-count, and the categories are wrong.

✓ STRONG / MODEL ANSWER

Biodiversity is the variety of all living things on Earth. Its three categories are genetic diversity (the variety of genes within a species), species diversity (the variety of species in an area) and ecosystem diversity (the variety of ecosystems in a region).

✗ WEAK ANSWER

Genetic diversity is about genes and species diversity is about species." — no within/between contrast and no examples, so at most 1 of 3.

✓ STRONG / MODEL ANSWER

Genetic diversity is the variety of genes (alleles) within a single species or population — for example, variation in coat colour among a mob of eastern grey kangaroos. Species diversity is the variety of different species in an area — for example, 40 bird species recorded in one woodland survey. The key difference is scale: genetic diversity is variation inside one species, whereas species diversity is variation across many species.

✗ WEAK ANSWER

They have the same diversity because both have 5 species." — confuses richness with diversity and ignores evenness entirely.

✓ STRONG / MODEL ANSWER

Both plots have the same species richness (5 species). However, species diversity depends on both richness and evenness. Plot A has high evenness — the 50 individuals are spread evenly (10 per species). Plot B is heavily dominated by one species (46 of 50 individuals), so its evenness is very low. Because Plot A combines the same richness with much higher evenness, Plot A has the greater species diversity.